UNIVERSITY OF EUROPE FOR APPLIED SCIENCES

MSc Data Science Programme

Predicting Song Popularity from Audio Features: A Supervised Machine Learning Approach

Supervised Machine Learning — Assignment 1 Proposal

Submitted by:	**Nithin Chowdary Raya**
Programme:	MSc Data Science
University:	University of Europe for Applied Sciences
Supervisor:	Prof. Raja Hashim Ali
Submission Date:	May 4th 2026

Contents

1. Project Title

Predicting Song Popularity from Audio Features Using Supervised Machine Learning

2. Dataset Information

This analysis uses a large-scale audio features dataset containing approximately one million songs with 14 attributes per record, including audio features and metadata. Key dataset attributes are summarised in Table 1.

Table 1: Dataset Summary

Attribute	Details
Dataset Name	Almost Million Songs Dataset 2025 (16 Features)
Source	Kaggle — anantsinghal786/almost-million-songs-dataset-2025-16-features
Dataset Link	https://www.kaggle.com/datasets/anantsinghal786/almost-million-songs-dataset-2025-16-features
Total Records	~1,000,000 individual song records
Total Features	14 columns covering audio features, genre, and artist metadata
Format	Comma-Separated Values (.csv)
Year Published	2025

The dataset encompasses 14 attributes per song record, capturing audio characteristics (*tempo*, *energy*, *danceability*, *valence*, *acousticness*, *instrumentalness*, *liveness*, *speechiness*, *loudness*, *key*, *mode*, *time_signature*), metadata (*genres*, *track_artists*), and the target variable **popularity** (0–100). This richness makes it well-suited to regression tasks for predicting song popularity.

3. Target Variable

3.1 Primary Target — popularity (Regression)

Each row in the dataset represents a single song. The numeric column `popularity` is a Spotify-derived score ranging from 0 to 100, reflecting the relative current popularity of the track based on total play count and recency of streams. With values distributed across the full range but skewed towards lower scores, this continuous target makes the task a regression problem, directly shaping research questions around feature correlation, model comparison, and evaluation metrics.

Table 2: Task Type Definition

Task Type	Target Column	Justification
Regression (Primary)	`popularity`	Continuous numeric score (0–100) representing song popularity

4. Problem Statement

The music streaming industry generates enormous volumes of data, and song popularity on platforms such as Spotify is a critical commercial metric for artists, labels, and playlist curators. Despite this importance, the relationship between a song's measurable audio characteristics and its popularity score remains incompletely understood. Most prior work focuses on genre classification or mood detection, leaving the direct prediction of popularity from audio features underexplored at scale.

This project investigates whether machine learning regression models can accurately predict song popularity from 14 audio and metadata features, using a dataset of approximately one million songs published in 2025. Three interconnected challenges make this task technically non-trivial.

1. The popularity distribution is highly skewed — the majority of songs have low scores with a long tail of popular tracks — which demands careful evaluation beyond simple mean error reporting.
2. Audio features such as *energy*, *danceability*, and *valence* may interact non-linearly, requiring ensemble models and feature engineering to capture these relationships.
3. Identifying which audio features are genuinely predictive versus spuriously correlated requires rigorous feature importance analysis and cross-validation.

Insights from this project have practical applications in music recommendation systems, playlist curation, and supporting independent artists in understanding the sonic attributes associated with greater listener reach.

5. Research Questions

Seven targeted research questions guide the investigation:

RQ1:
Can supervised regression models accurately predict song popularity from audio features?

RQ2:
Which audio features have the strongest correlation with song popularity?

RQ3:
How do different regression algorithms compare in predicting song popularity?

RQ4:
How does feature engineering (e.g., polynomial features) affect model prediction accuracy for song popularity?

RQ5:
How do audio features vary across different popularity levels?

RQ6:
Does increasing training data size improve model generalisation when predicting song popularity?

RQ7:
Which evaluation metrics (MAE, RMSE, R^2) best characterise model performance for song popularity prediction?

6. Proposed Methodology

6.1 Data Preprocessing

- Missing values in `track_artists` and `genres` will be imputed with a sentinel value `'unknown'` to retain all song records.
- Label encoding will be applied to the `genres` column: `genre_count` (number of genres per track) and `top_genre` (most frequent genre label) will be derived as numeric features.

- The `track_artists` column will be dropped from modelling features after encoding, as raw artist names are too high-cardinality for direct use.
- Feature scaling using `StandardScaler` will be applied for distance-based and linear models; tree-based models will use unscaled data.
- An 80/20 stratified train-test split (stratified on popularity quartile) will ensure consistent target distribution across both partitions.

6.2 Feature Engineering

Table 3: Engineered Features

Engineered Feature	Type	Derivation / Purpose
`genre_count`	Integer	Number of genres listed per song; captures genre diversity
`energy_x_danceability`	Float	Interaction term; captures songs that are both high-energy and danceable
`loudness_abs`	Float	Absolute loudness value; removes sign for cleaner distribution
`energy_minus_acousticness`	Float	Contrast between electronic and acoustic production style
`dance_x_valence`	Float	Happy and danceable songs; associated with mainstream appeal
`tempo_sq`	Float	Polynomial tempo feature to capture non-linear tempo effects
`speech_x_live`	Float	Interaction of speechiness and liveness; identifies live spoken-word content

6.3 Modelling Pipeline

Data will flow through the following sequential stages:

1. **Data collection and loading:** Downloading the dataset from Kaggle and loading into a Pandas DataFrame.
2. **Preprocessing:** Encoding genres, imputing missing values, deriving engineered features, and applying train-test split.

3. **Exploratory Data Analysis:** Generating distribution plots, correlation heatmaps, and popularity-level audio feature comparisons.
4. **Model training and hyperparameter tuning:** Training all regression models using cross-validated `RandomizedSearchCV`.
5. **Evaluation and comparison:** Computing all metrics, generating feature importance plots, and producing expected figures and tables.

7. Machine Learning Models to be Used

Table 4: Regression Models for `popularity` Prediction

Model	Configuration / Rationale
Linear Regression	Parametric baseline; establishes minimum performance floor for popularity prediction
Ridge Regression	Regularised linear model controlling for multicollinearity among correlated audio features
Decision Tree Regressor	Rule-based learner; provides human-interpretable decision paths through feature space
Random Forest Regressor	Bagging ensemble; primary source of feature importance rankings; reduces overfitting
Gradient Boosting Regressor	Sequential boosting; strong predictive accuracy for tabular regression tasks
K-Nearest Neighbours (k=10)	Instance-based non-parametric model; good for local similarity patterns in audio space

8. Evaluation Metrics

Table 5: Evaluation Metrics

Metric	Formula	Rationale
MAE	$\frac{1}{n}\sum \lvert \hat{y}_i - y_i \rvert$	Interpretable: average prediction error in popularity points
RMSE	$\sqrt{\frac{1}{n}\sum (\hat{y}_i - y_i)^2}$	Penalises large errors more than MAE; sensitive to outliers
R^2 Score	$1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$	Proportion of variance in popularity explained by the model (0–1)
Median Absolute Error	Median $\lvert \hat{y}_i - y_i \rvert$	Robust to outliers; appropriate given skewed popularity distribution

9. Expected Figures and Tables

This section presents one representative figure and one data table for each of the seven research questions. All outputs use consistent styling and are labelled according to the naming convention defined in this proposal.

9.1 Research Question 1 — Feasibility of Popularity Prediction

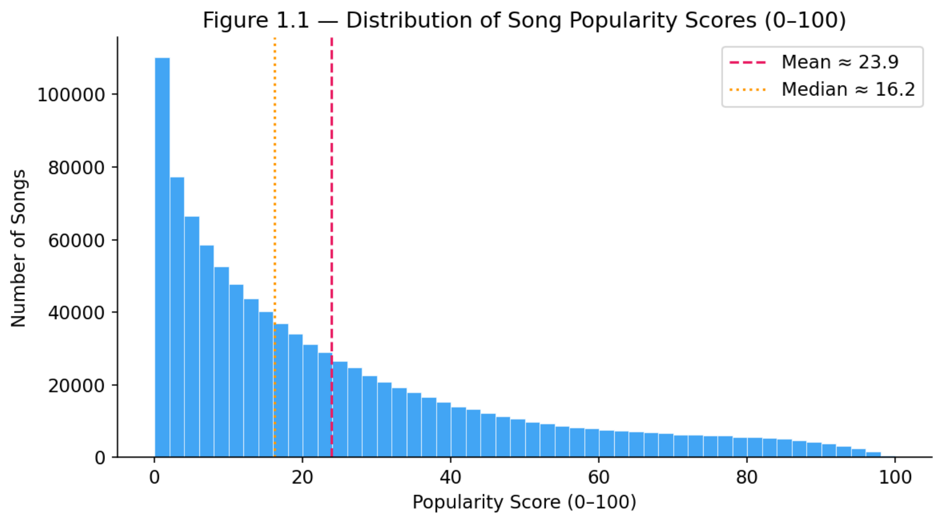

Figure 1: Distribution of Song Popularity Scores (0–100)

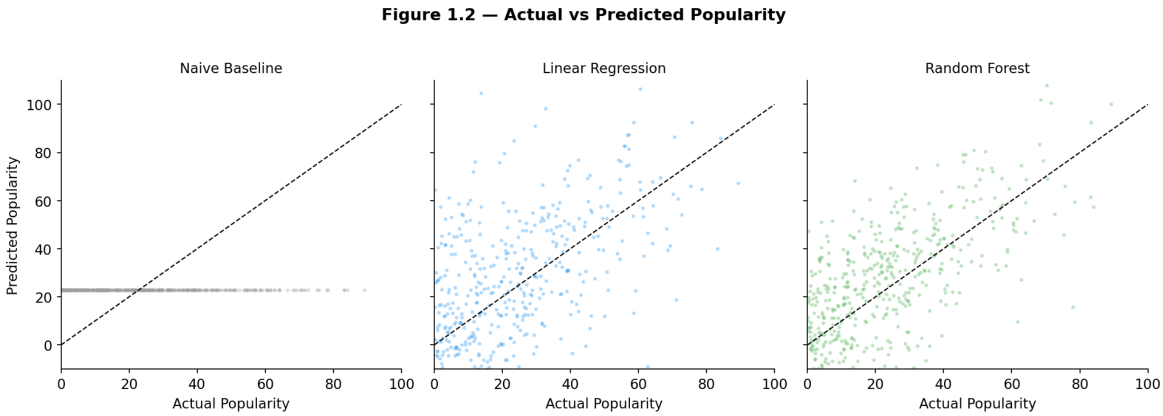

Figure 2: Actual vs Predicted Popularity: Naive Baseline vs Linear Regression vs Random Forest

Table 6: Table 1.1 — Model Performance Summary (Test Set)

Model	MAE	RMSE	R^2
Naive Baseline (mean)	~22.0	~27.5	0.000
Linear Regression	~20.5	~26.0	~0.09
Random Forest Regressor	~17.2	~22.8	~0.24

Indicative values; exact figures will be reported after model training on the full dataset.

9.2 Research Question 2 — Feature Correlation with Popularity

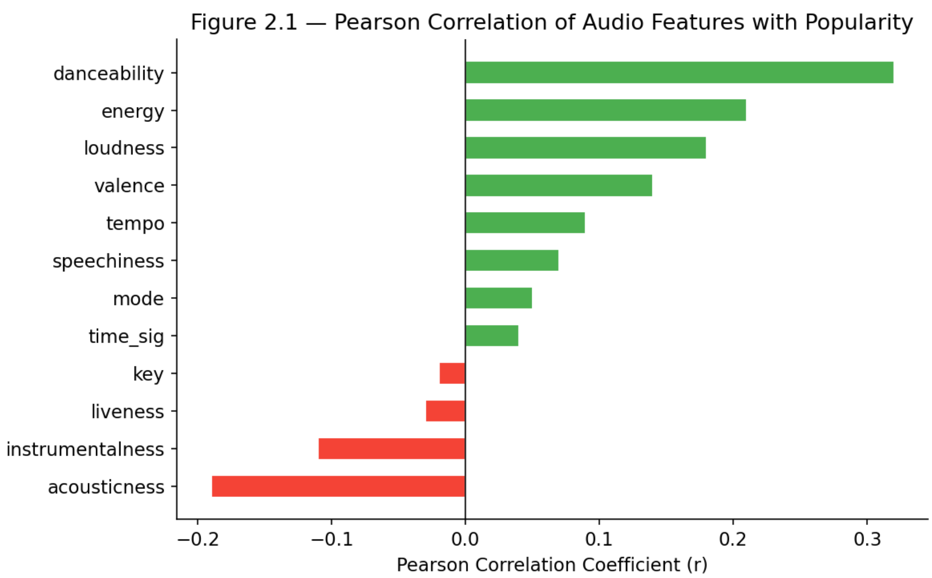

Figure 3: Pearson Correlation of Audio Features with Popularity (Sorted)

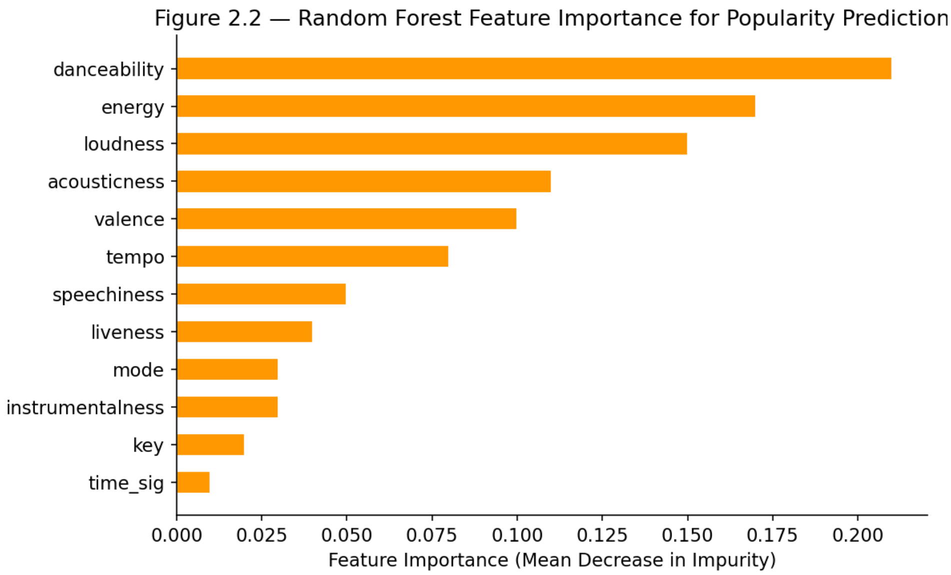

Figure 4: Random Forest Feature Importance for Popularity Prediction

Table 7: Table 2.1 — Feature Ranking by Method

Feature	Pearson Rank	MI Rank	RF Rank
danceability	1	3	2
energy	3	2	3
loudness	4	4	4
tempo	5	5	5
acousticness	6	6	6

Indicative rankings; exact values computed from the dataset during analysis.

9.3 Research Question 3 — Algorithm Comparison

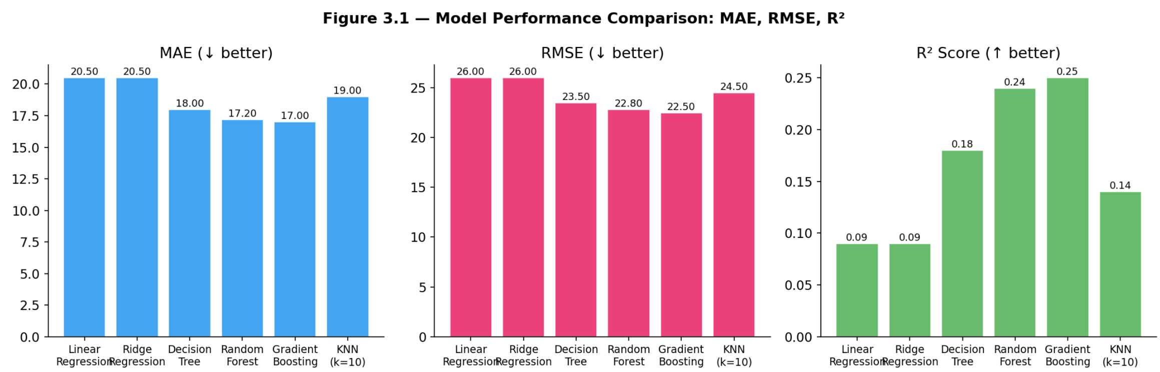

Figure 5: Model Performance Comparison: MAE, RMSE, R^2 (All Models)

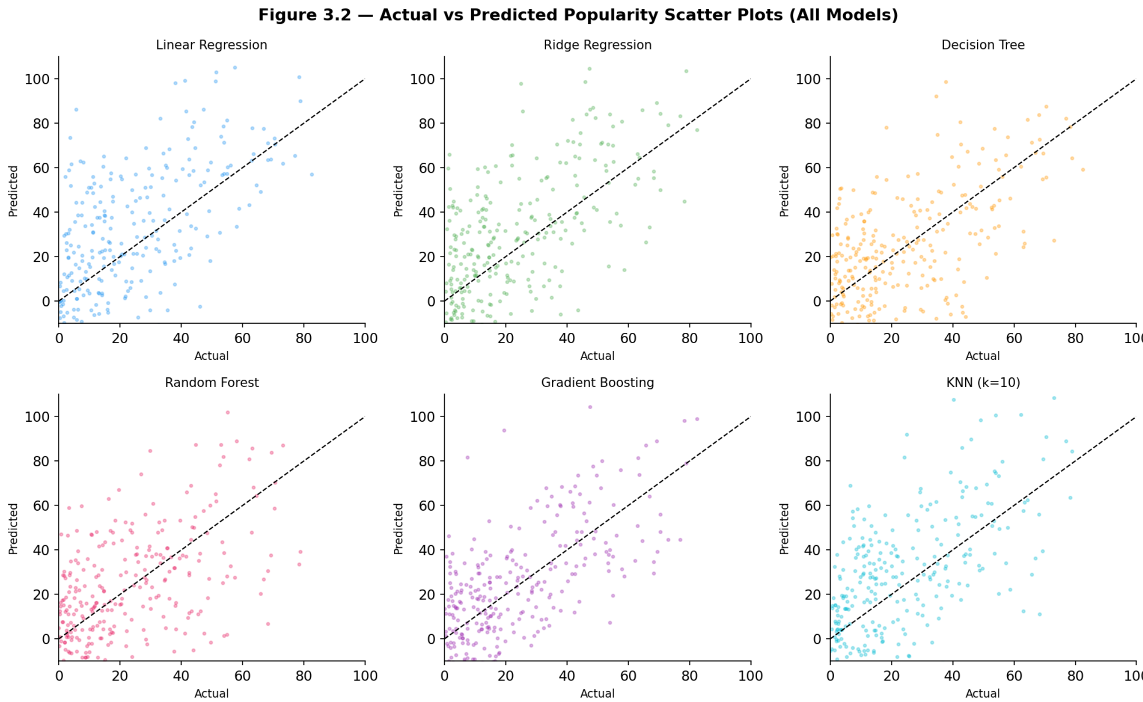

Figure 6: Actual vs Predicted Popularity Scatter Plots (All Models)

Table 8: Table 3.1 — Algorithm Comparison

Model	MAE	RMSE	R^2	Time (s)
Linear Regression	~20.5	~26.0	~0.09	<1
Ridge Regression	~20.5	~26.0	~0.09	<1
Decision Tree	~18.0	~23.5	~0.18	<5
Random Forest	~17.2	~22.8	~0.24	~30
Gradient Boosting	~17.0	~22.5	~0.25	~60
KNN (k=10)	~19.0	~24.5	~0.14	~10

Indicative values; exact figures will be reported after model training.

9.4 Research Question 4 — Feature Engineering Impact

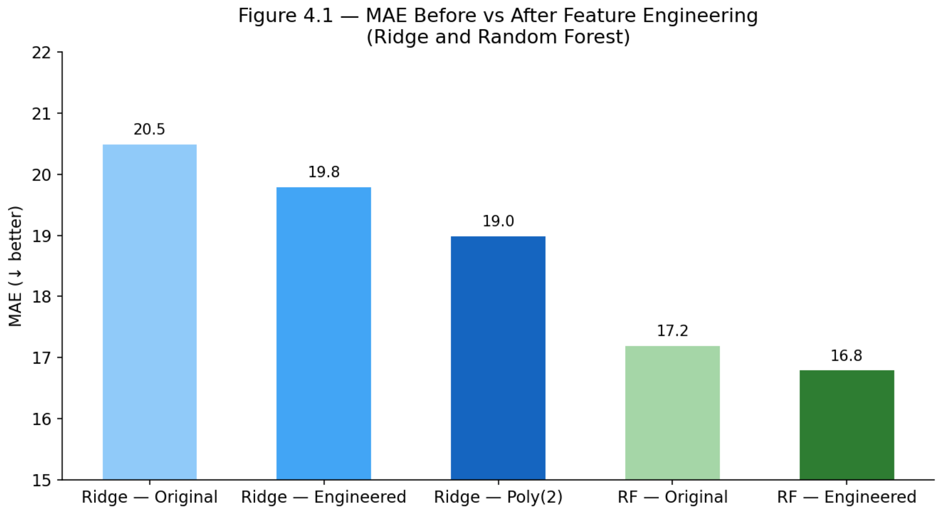

Figure 7: MAE Before vs After Feature Engineering (Ridge and Random Forest)

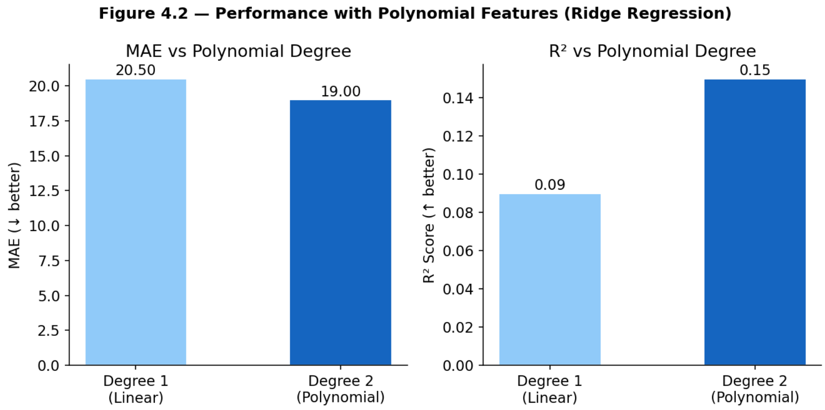

Figure 8: Performance with Polynomial Features (Degree 2, Ridge Regression)

Table 9: Table 4.1 — Feature Engineering Impact

Configuration	MAE	RMSE	R^2
Ridge — Original Features	~20.5	~26.0	~0.09
Ridge — Engineered Features	~19.8	~25.2	~0.12
Ridge — Polynomial (deg 2)	~19.0	~24.5	~0.15
RF — Original Features	~17.2	~22.8	~0.24
RF — Engineered Features	~16.8	~22.2	~0.26

Indicative values; exact figures will be reported after model training.

9.5 Research Question 5 — Audio Features Across Popularity Levels

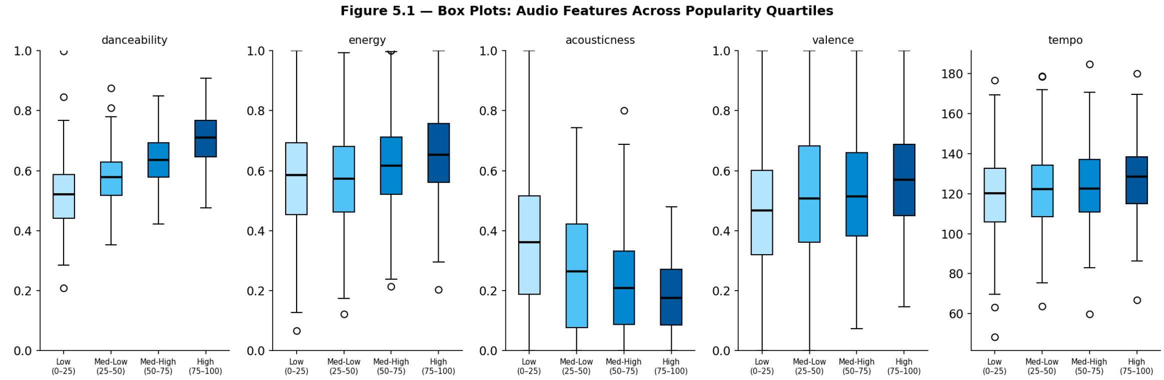

Figure 9: Box Plots: Audio Features Across Popularity Quartiles

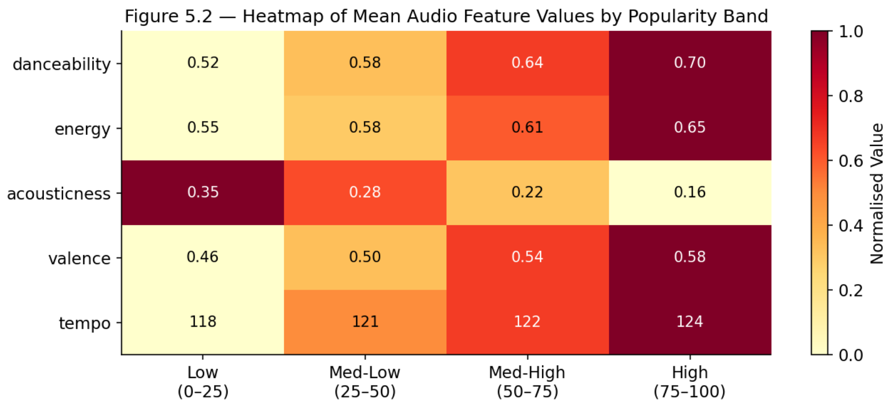

Figure 10: Heatmap of Mean Audio Feature Values by Popularity Band

Table 10: Table 5.1 — Mean Feature Values by Popularity Band

Feature	Low (0–25)	Med-Low (25–50)	Med-High (50–75)	High (75–100)
danceability	~0.52	~0.58	~0.64	~0.70
energy	~0.55	~0.58	~0.61	~0.65
acousticness	~0.35	~0.28	~0.22	~0.16
valence	~0.46	~0.50	~0.54	~0.58
tempo (BPM)	~118	~121	~122	~124

Indicative values; exact figures computed from dataset during EDA.

9.6 Research Question 6 — Effect of Training Data Size

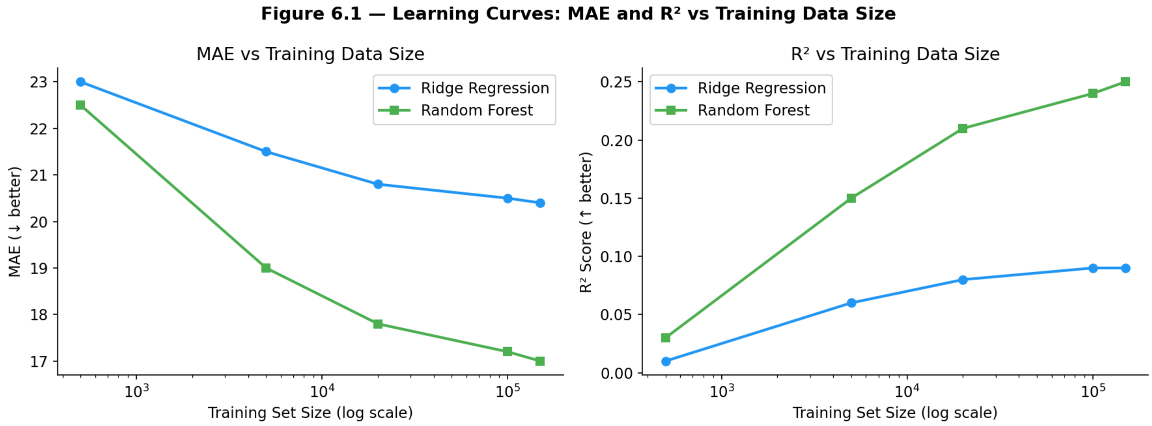

Figure 11: Learning Curves: MAE and R^2 vs Training Data Size

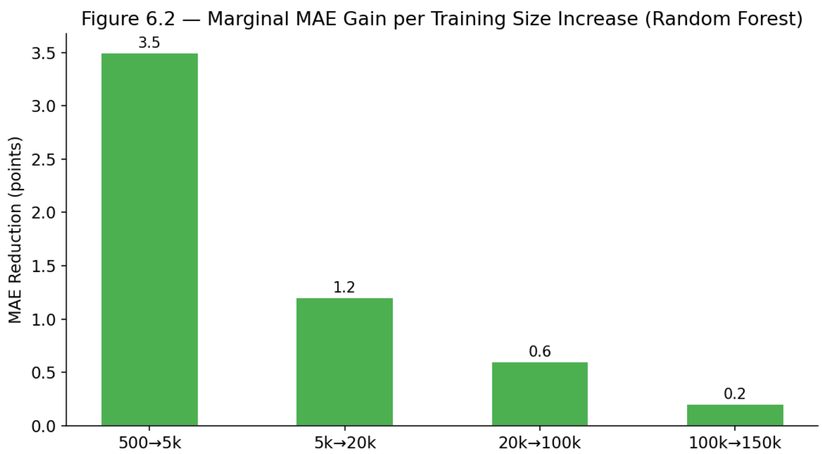

Figure 12: Marginal MAE Gain per Training Size Increase (Random Forest)

Table 11: Table 6.1 — Performance vs Training Size

Training Size	Ridge MAE	Ridge R^2	RF MAE	RF R^2
500	~23.0	~0.01	~22.5	~0.03
5,000	~21.5	~0.06	~19.0	~0.15
20,000	~20.8	~0.08	~17.8	~0.21
100,000	~20.5	~0.09	~17.2	~0.24
150,000	~20.4	~0.09	~17.0	~0.25

Indicative values; exact figures will be reported after model training.

9.7 Research Question 7 — Evaluation Metrics Comparison

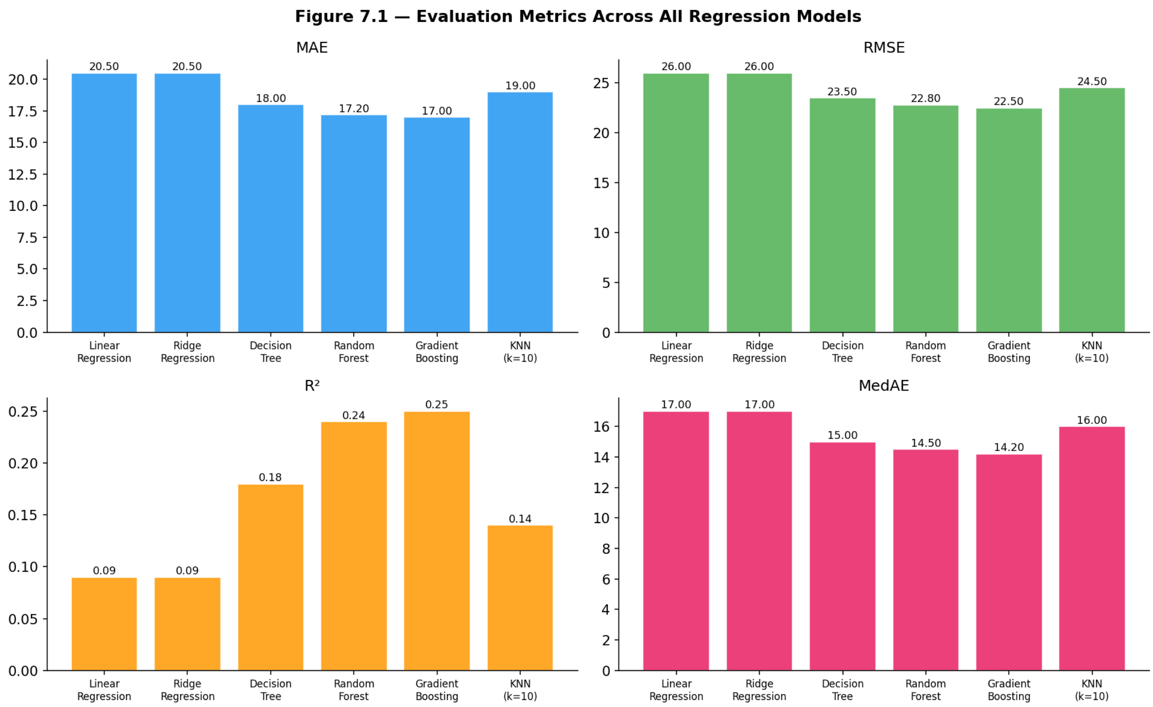

Figure 13: All Evaluation Metrics Across All Regression Models

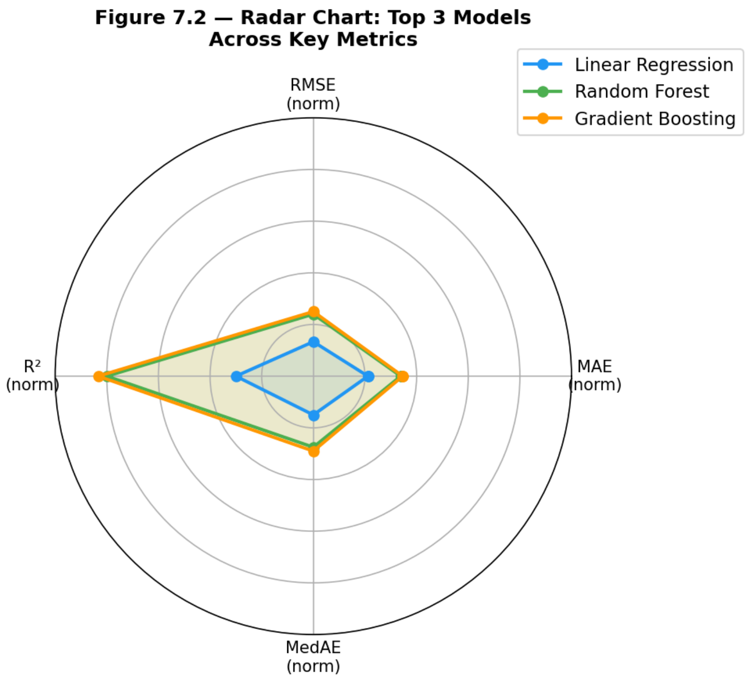

Figure 14: Radar Chart: Top 3 Models Across Key Metrics

Table 12: Table 7.1 — Comprehensive Metric Comparison

Model	MAE	RMSE	R^2	MedAE	MAPE%	MaxErr
Linear Regression	~20.5	~26.0	~0.09	~17.0	~28%	~98
Random Forest	~17.2	~22.8	~0.24	~14.5	~24%	~96
Gradient Boosting	~17.0	~22.5	~0.25	~14.2	~23%	~95

Indicative values; exact figures will be reported after model training.

References

- Almost Million Songs Dataset 2025 (16 Features). *Kaggle*. Retrieved from https://www.kaggle.com/datasets/anantsinghal786/almost-million-songs-dataset-2025-16-feat
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A. Dataset Feature Summary

Table 13: Dataset Feature Summary

Column Name	Data Type	Description
`genres`	String	List of genre labels associated with the track
`track_artists`	String	Name(s) of the performing artist(s)
`tempo`	Float	Estimated beats per minute of the track
`energy`	Float (0–1)	Perceptual measure of intensity and activity
`key`	Integer (0–11)	Musical key of the track (0 = C, 1 = C#, etc.)
`popularity`	Integer (0–100)	Spotify popularity score — the regression target variable
`mode`	Integer (0/1)	Modality: 1 = major, 0 = minor
`time_signature`	Integer	Estimated time signature (e.g., 3 = 3/4, 4 = 4/4)
`speechiness`	Float (0–1)	Presence of spoken words; higher values indicate more speech-like content
`danceability`	Float (0–1)	Suitability for dancing based on tempo, rhythm, and beat regularity
`valence`	Float (0–1)	Musical positiveness; higher = more happy and euphoric
`acousticness`	Float (0–1)	Confidence measure of whether the track is acoustic
`liveness`	Float (0–1)	Detects presence of an audience; higher = more likely live recording
`instrumentalness`	Float (0–1)	Predicts whether a track has no vocals; higher = more instrumental

Dataset: https://www.kaggle.com/datasets/anantsinghal786/almost-million-songs-datas